



A HYBRID RAG SYSTEM WITH GRAPH BASED CONTEXT RETRIEVAL

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THE PROBLEM

- Academic LLMs produce hallucinated answers due to lack of citation awareness
- Users cannot easily verify claims, reducing trust in AI-assisted research.

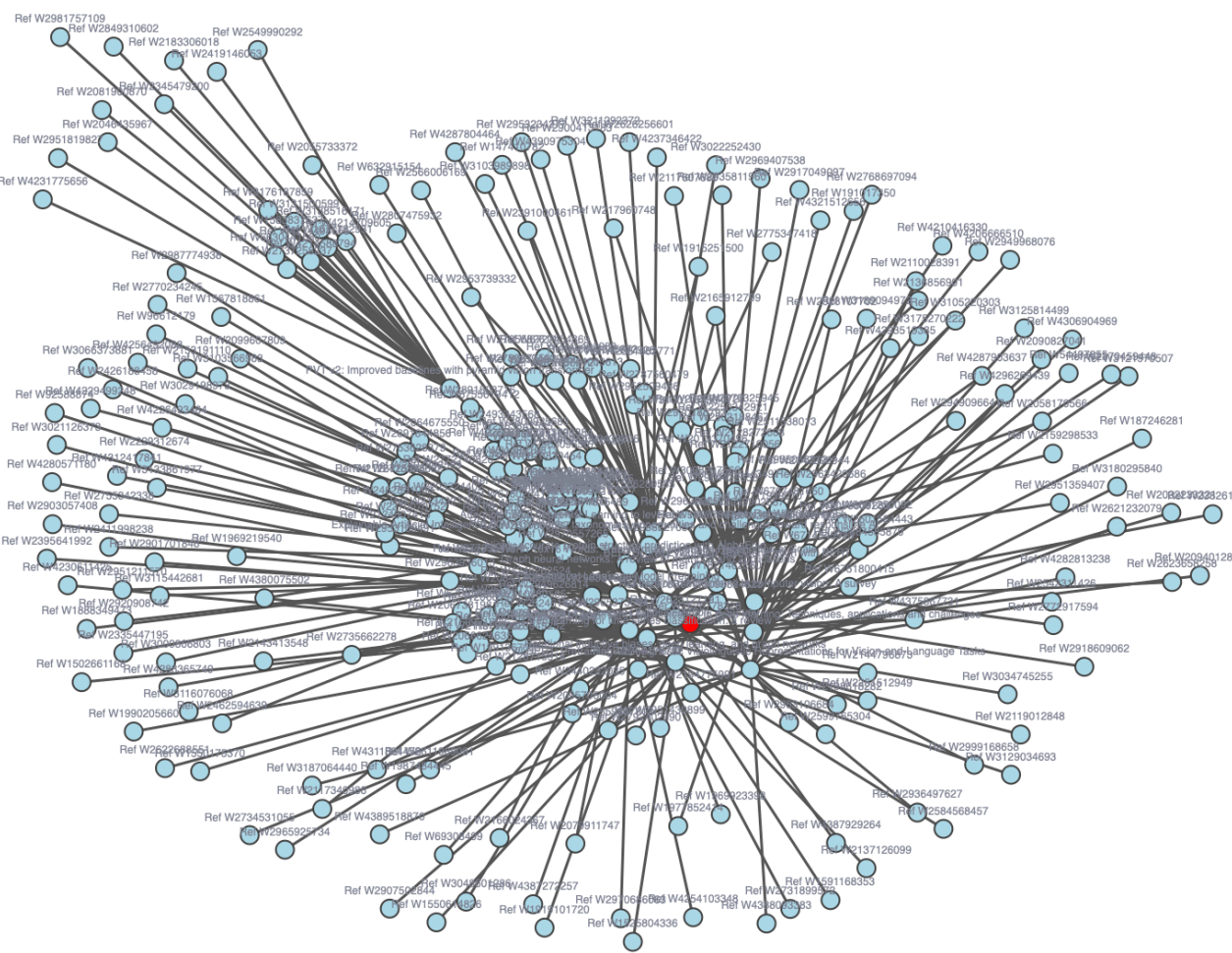
Question: How can we build a retrieval system that not only returns relevant content, but also explains its answers by tracing verified citation paths in academic literature?

SOLUTION

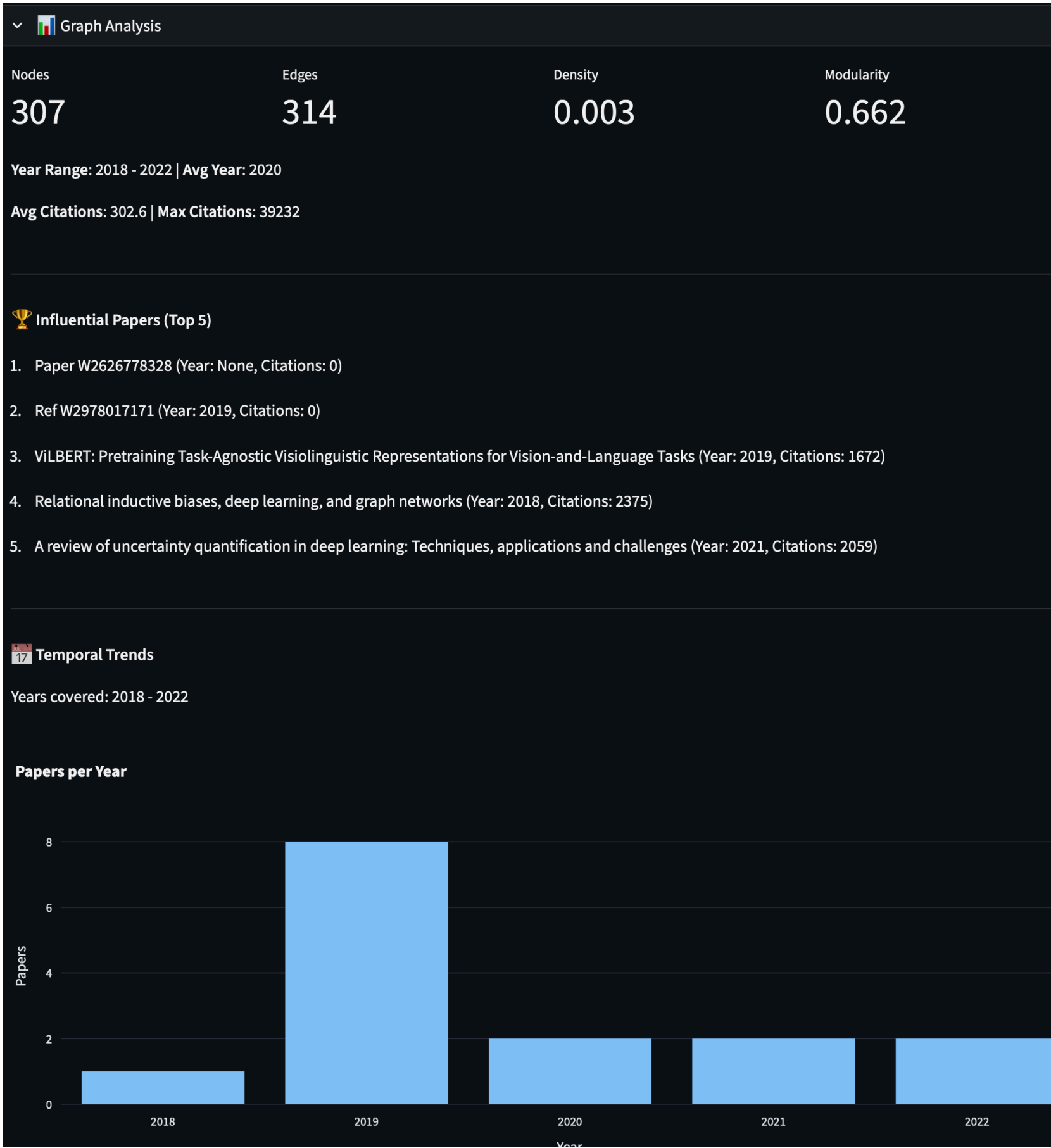
Hybrid Graph-RAG Academic Assistant

- Parses PDFs & builds citation graphs →
- Dual retrieval with vector + citation paths →
- Explains answers with trails & confidence scores →
- UI for queries & visualization

CITATION GRAPH & ANALYSIS



The **graph analysis module** builds a citation network from ingested papers, computes structural metrics, and highlights the most influential nodes and communities. It surfaces top-cited or central papers, tracks temporal trends in publications, and visualizes how ideas propagate across years, helping users quickly identify key works and evolving research fronts.

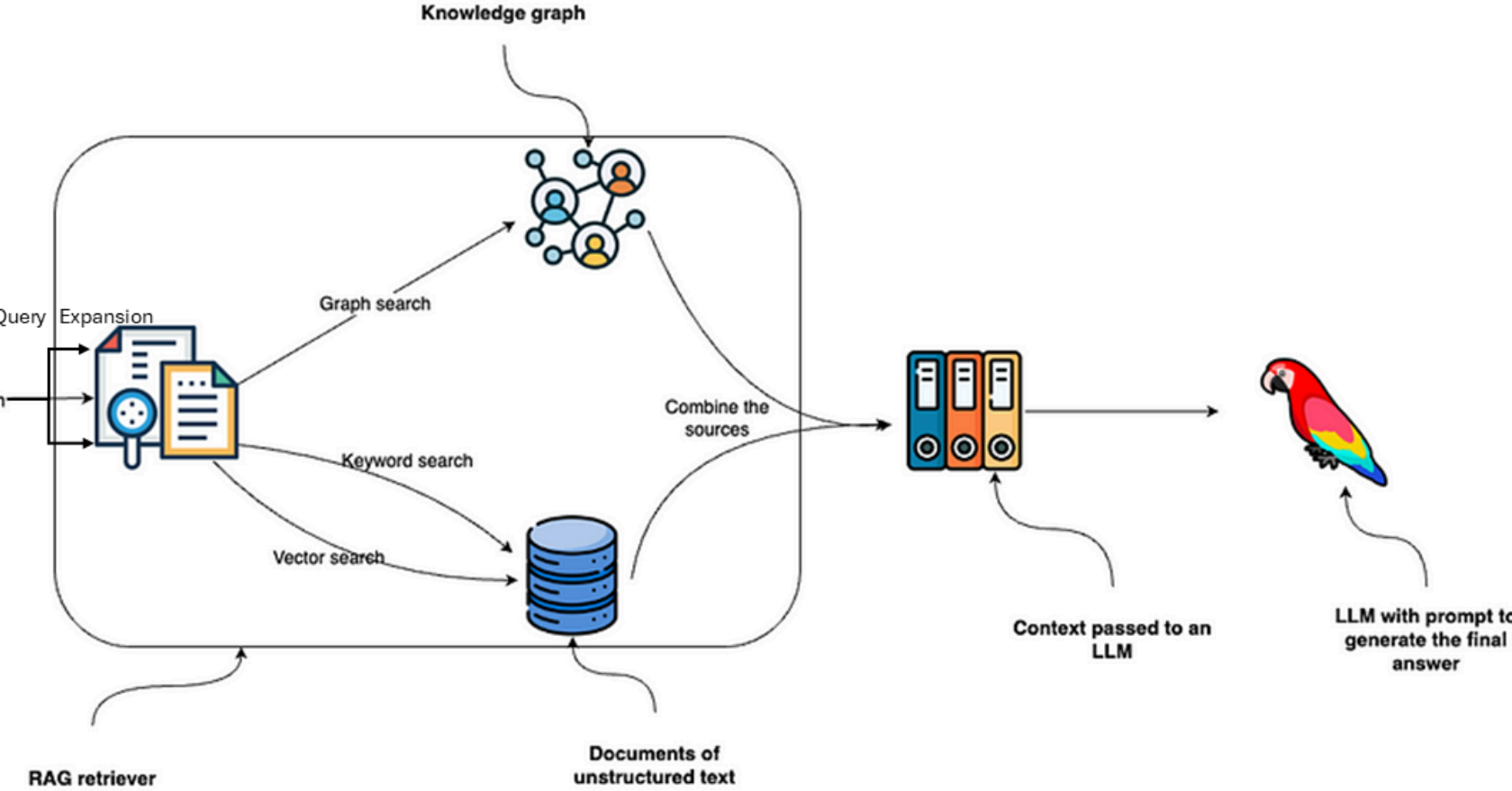
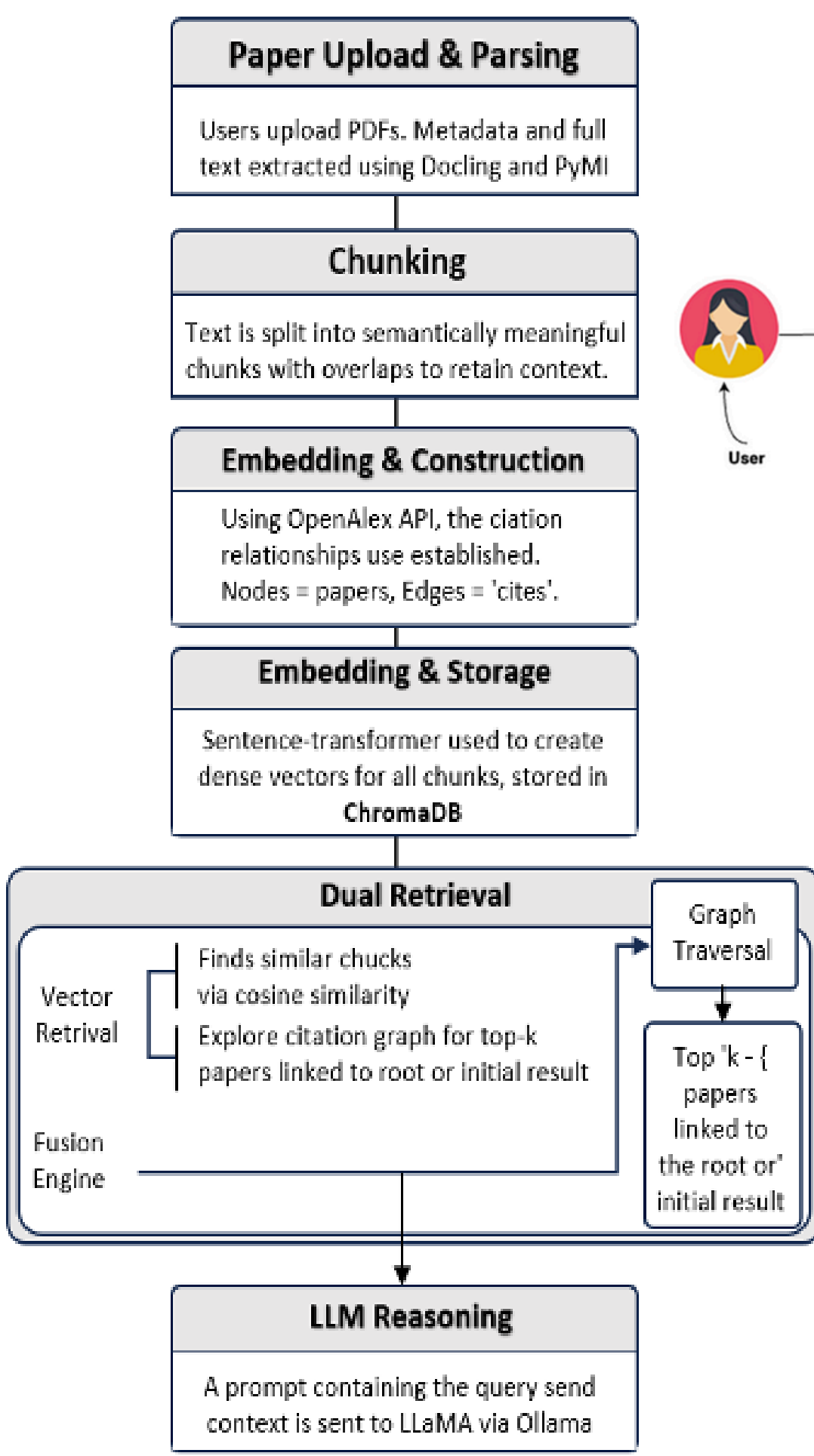


Citation Path: Ref W2978017171 (2019, citations:0) —cited_root (w=45.38)→ Paper W2626778328 (citations:0) → ViLBERT: Pretraining Task-Agnostic Visiolinguistic Representations for Vision-and-Language Tasks (2019, citations:1672) —cited_root (w=16.72)→ Paper W2626778328 (citations:0) → An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale (2020, citations:20679) —cited_root (w=206.79)→ Paper W2626778328 (citations:0)

CONCLUSION

- Hybrid Graph-RAG** delivers more accurate, transparent, and verifiable academic answers.
- Citation networks** add structure that semantic search alone cannot capture.
- The system provides clear source mapping and a simple UI for querying & visualization.
- A promising direction for building trustworthy academic AI tools.**

ARCHITECTURE & SYSTEM OVERVIEW



User Input: A user submits a question to the system.

Query Expansion: The LLM expands the user query to retrieve more contextual information/chunks.

RAG Retriever: This module performs the following parallel searches to find relevant information:

- Vector Search** in the **Document Store** (unstructured text).
- Graph Search** in the **Knowledge Graph** (structured relations).

Context Engineering: The retrieved chunks are re-ranked based on relevance scores and semantic similarity.

Combine Sources(Fusion): The results from the searches are merged to create a comprehensive context, reducing hallucinations while improving trust.

Final Answer: The Context is passed to an LLM (Large Language Model) with a prompt to generate the final answer.

EXPLAINABILITY & TRACEABILITY

Citation Backed Reasoning: Every response links to verified papers and citation paths.

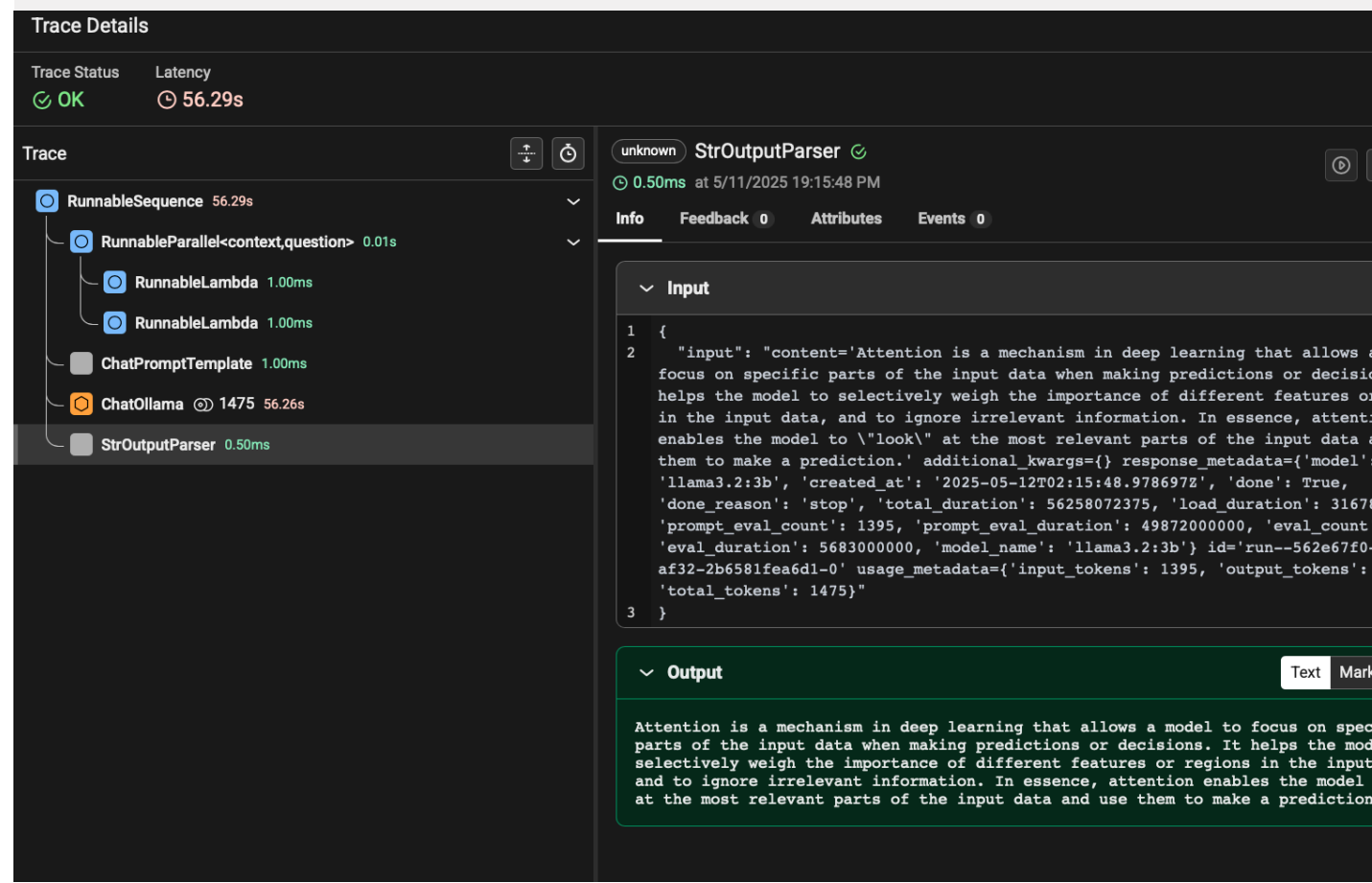
Chunk-Level Attribution: Highlights the exact text segments that support each answer.

Graph-Based Traceability: Visualizes relationships among retrieved papers for easy traceability.

Confidence Scores: Combines semantic relevance and citation strength

End-to-End LLM Request Tracing

- Powered by Phoenix for monitoring LLM interactions.
- Every call is logged with inputs, outputs, and execution metadata for full transparency



REFERENCES

- QGCRerank* - <https://arxiv.org/pdf/2411.08724>
- RAG* - <https://arxiv.org/pdf/2005.11401>
- Hybrid RAG* - <https://arxiv.org/pdf/2408.04948>